7. Version Control-GIT: GIT Features, 3-Tree Architecture, GIT – Clone /Commit / Push.

Step 1:Install Git (if not installed)

```
sudo apt update
sudo apt install git -y
```

Check:

```
git -version
```

```
git config --list
```

Step 2 – Configure Git (first time only)

```
git config --global user.name "Dr-Yasmeen"
git config --global user.email "dr.yasmeen.aashu@gmail.com"

git config --global --list
```

STEP 3: Create or open a Git repository

I)Option A: Initialize a new repository

```
mkdir my_project
cd my_project
git init
```

ii) Option B: Clone an existing repository--

```
git clone https://github.com/username/repository.git
cd repository
```

4. Check file status

```
git status
```

5.Add files (stage)

Add all files:

```
git add .
```

Or a specific file:

```
git add paper.txt
```

6. Commit changes

```
git commit -m "created file f1"
```

7. Connect to remote repository (GitHub/GitLab)

```
git remote add origin https://github.com/username/repository.git
```

Verify:

```
git remote -v
```

8.Push to remote repository

First push:

```
git branch -M main
git push -u origin main
```

(or `master` if your branch is master)

git pull origin main --allow-unrelated-histories --no-rebase

Way to Connect Ubuntu to GitHub (HTTPS + Token)

Step 1: Generate Token on GitHub

1. Go to **GitHub → Profile → Settings**
2. Click **Developer settings**

3. Click **Personal access tokens**
4. Click **Tokens (classic)**
5. Click **Generate new token (classic)**

Select:

repo (very important)

Click **Generate token**

Copy the token (you won't see it again).

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

go to terminal--

```
git push -u origin main
```

When it asks:

Username: Dr-Yasmeen

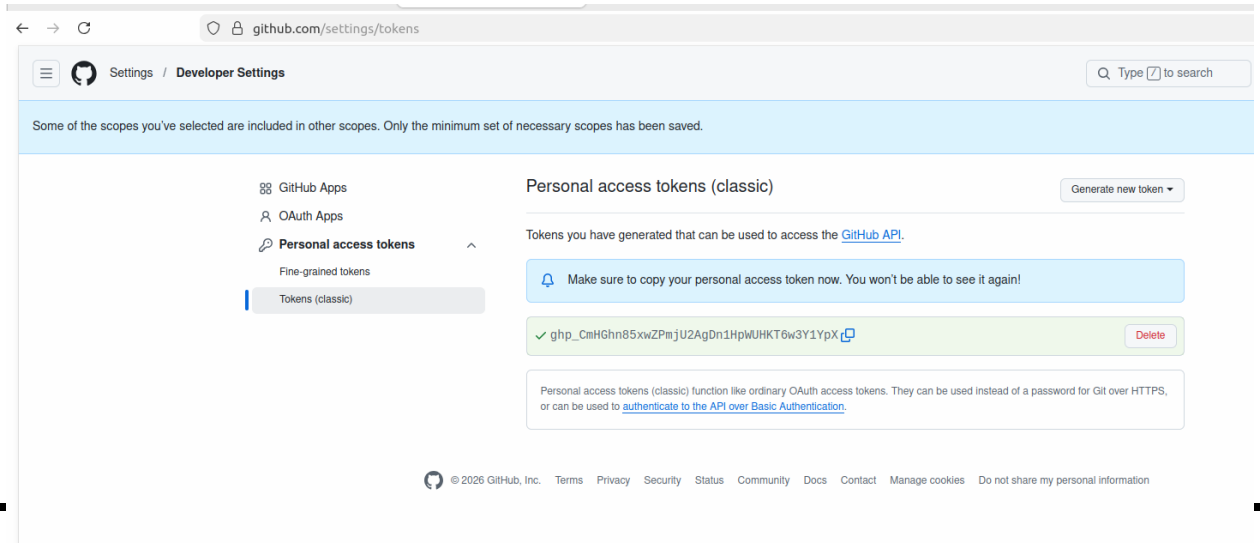
When it asks:

Password:

Paste the TOKEN (NOT your GitHub password)

You won't see characters while typing — that's normal.

Press Enter.



Next pushes:

```
git push
```

8. Pull latest changes

```
git pull
```

```
git log
git log --oneline
```

BRANCH-(OPTIONAL)

git branch branch name

git checkout -b branch-name

git branch— check current branch-name

git push -u origin branch-name ==-u sets upstream so next time you can just use:

git push --set-upstream origin dep

git push -u origin dep